

Breadth First Search (BFS)

1 Introduction

Breadth First Search (BFS) is a graph traversal algorithm that explores vertices level by level. It visits all neighbors of a vertex before moving to the next level.

BFS uses a **queue (FIFO)** data structure and is widely used in:

- Shortest path in unweighted graphs
- Level order traversal
- Network broadcasting

2 BFS Algorithm (Using Queue)

Algorithm 1 Breadth First Search

Require: Graph $G(V, E)$, starting vertex s

Ensure: BFS traversal order

```
1: Create an empty queue  $Q$ 
2: Mark all vertices as unvisited
3: Enqueue  $s$  into  $Q$ 
4: Mark  $s$  as visited
5: while  $Q$  is not empty do
6:    $v \leftarrow$  dequeue from  $Q$ 
7:   Process  $v$ 
8:   for each adjacent vertex  $u$  of  $v$  do
9:     if  $u$  is not visited then
10:      Mark  $u$  as visited
11:      Enqueue  $u$  into  $Q$ 
12:    end if
13:  end for
14: end while
```

3 Graph Example

Consider a graph with 8 vertices:

$$V = \{A, B, C, D, E, F, G, H\}$$

Adjacency list:

$A : B, C$

$B : D, E$

$C : F$

$D : H$

$E : -$

$F : G$

$G : -$

$H : -$

BFS traversal starts from vertex A .

4 BFS Traversal Order

The BFS traversal order is:

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow H \rightarrow G$$

5 BFS Queue Iterations

Iteration 1: Enqueue A

A

Front

Iteration 2: Dequeue A, Enqueue B, C

B

C

Front

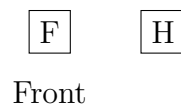
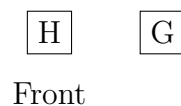
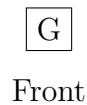
Iteration 3: Dequeue B, Enqueue D, E

C

D

E

Front

Iteration 4: Dequeue C, Enqueue F**Iteration 5: Dequeue D, Enqueue H****Iteration 6: Dequeue E****Iteration 7: Dequeue F, Enqueue G****Iteration 8: Dequeue H****Iteration 9: Dequeue G (Queue Empty)**

BFS traversal is complete.

6 Time and Space Complexity

- **Time Complexity:** $O(V + E)$
- **Space Complexity:** $O(V)$

7 Conclusion

Breadth First Search explores graphs in level order and guarantees the shortest path in unweighted graphs. It is widely used in routing, network analysis, and graph-based problem solving.